

1. Materials to Use

- Materials:
- Aerogels
- 3D-printed structures using local Martian soil
- Metal

2. Air Supply

- Electrolysis of Martian ice to separate oxygen and hydrogen.
- MOXIE (Mars Oxygen In-Situ Resource Utilization Experiment) to convert CO₂ from the Martian atmosphere into oxygen.
- Pressure-regulated living quarters with air recycling and CO₂ scrubbers.

3. Water Supply

- Extract water from underground Martian ice deposits.
- Filtration and recycling systems (closed-loop system) to reuse water from showers, washing, and urine.
- Hydroponics systems to provide additional water through transpiration.

4. Food Supply

- Hydroponic and aeroponic farming using nutrient-rich solutions.
- Lab-grown meat for protein.
- Food storage from Earth for backup.

5. Crew Mobility

- Transport system within the base.
- rovers for surface exploration.
- Personal suits with exoskeleton support for heavy work.

6. Radiation and Environmental Protection

- Thick domelike shielding .
- Underground living quarters for additional protection.

7. Power Sources

- Nuclear fission reactors for continuous power.
- Solar panels with dust-resistant coating.

8. Mental and Physical Health

- Artificial gravity rings (centrifuge-style) for muscle and bone health.
- Social spaces for recreation and bonding.
- Exercise facilities (gyms, climbing walls, etc.).

9. Living and Recreational Spaces

- Modular living pods with personalized interiors.
- Central greenhouse for psychological and physical health benefits.
- Sports/workout areas
- Entertainment hub with movies, games, and live music.

10. Communication with Earth

- High-gain antennas for fast data transmission.
- Laser-based communication for reduced lag time.

11. Additional Systems and Benefits

- Emergency escape pods to return to orbit.
- Robotic maintenance teams for repairing external structures.

